



Q1. What is a switch debouncer? How does it work?

Suresh

A1. Mechanical switches are one of the most common components used in electrical and electronics manual control circuits. When you press a mechanical switch, two metal parts come in contact with each other, and the current flows through the contacts. But sometimes the metal parts connect and disconnect several times before the actual stable connection is made. The same thing happens while releasing the switch button. The mechanical switch bounce or contact bounce, sometimes called chatter, is a common problem associated with mechanical switches and relays.

Note that each switch has its own characteristics regarding the bounce. For example, you may have two similar switches, but there is a high chance that each will bounce differently. These problems are very common in digital and microcontroller circuits associated with the mechanical switch. These chattering inputs are electrically cleansed with a switch debouncer circuit to prevent or reduce false trigger-

ing or multiple triggering.

One of the simplest solutions is adding a resistor and capacitor (RC) across the switch, as shown in Fig. 1. Here, the RC circuit acts as a filter to smooth out the output glitches. When the switch is open, the voltage across the capacitor remains zero. Initially, the capacitor charges through the resistors (R_p and R_d).

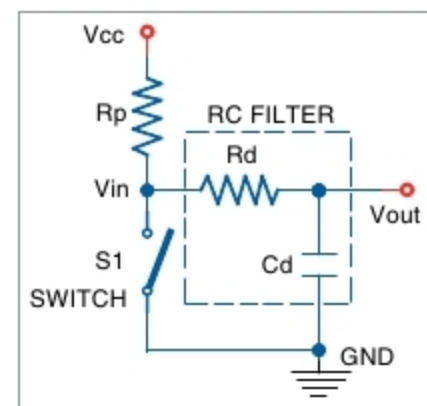


Fig. 1: Debounce circuit

When the switch is closed, the capacitor starts discharging through R_d , and hence the voltage output (V_{out}) becomes high. In the bouncing condition, the capacitor stops the voltage at the input voltage (V_{in}) until it reaches V_{cc} or ground, preventing false triggering.

Q2. Which are long-range wireless modules that can operate in the line-of-sight of up to one kilometre range? Kindly suggest some modules easily available in the market.

Pamarthi Kanakaraja

A2. Long-range wireless sensor modules are used in many applications, including asset tracking systems, community building security, robot wireless control, industrial remote control, telemetering, automatic data acquisition, vehicle keyless entry system, PC wireless networking, toys, and drones among others.

One such sensor is the nRF24L01 module (refer Fig. 2) from Nordic Semiconductor. For application, you may refer to 'Ultra-Low-Power Sensor Hub Using nRF24L01 Modules' DIY article available on <https://www.electronicsforu.com/electronics-projects/ultra-low-power-sensor-hub-using-nrf24l01>.

Some long-range wireless sensor modules easily available in the Indian market include nRF24L01, Zigbee XBee Pro S2C 802.15.4, HC-12, TS835 FPV 5.8G, and LoRa SX1278 module.

1. nRF24L01 module. Its operating range is up to one kilometre and it uses 2.4GHz global open ISM band.



Fig. 2: nRF24L01 module

2. Zigbee XBee Pro S2C 802.15.4 module. Its range for outdoor line-of-sight is up to 3.2km. The operating frequency is 2.4GHz for worldwide deployment.

3. HC-12 wireless serial port communication module. It has a range of up to one kilometre in open space. It works in the frequency range of 433.4-473.0MHz and supports up to a hundred communication channels.

4. TS835 FPV 5.8G 600MW 48CH (2-6S) wireless AV transmitter module. It has a range of 5km to 8km when used with a bigger gain antenna. The operating frequency is 5.8GHz.

5. LoRa SX1278 long-range RF wireless power mental module. It has a range of up to 5km. This module uses SX1278 IC from SEMTECH and works on 433MHz frequency.

There is also a long-range wireless module, the F8L10 series, which can work up to 11km.